29-03-2026

Questions and Follow-Ups!

If you have any questions at all relating to informatics, or want to follow up something from class, feel free to send me a message through AUCoding Admin.

Lesson: A Problem Session :D

Solutions to Select Homework Tasks:

The Great Revegetation

> **Practice**
>
> For those interested, this is a modification of the **two-colouring** problem.
>
> I recommend https://orac2.info/problem/117/ (King Arthur II) as follow up.

Evading Capture (AIO'19 T5)

> **Extension**
>
> The general technique used here is known as **state explosion**.

Today's Problems:

[Sharky Surfing]()

sharky.cpp

```cpp
#include<bits/stdc++.h>
using namespace std;
#define int long long
#define arr array

const int MAXN = 2e5+5;
int n, m, L, l[MAXN], r[MAXN], p[MAXN], v[MAXN];

void testcase(int tc) {
    cin >> n >> m >> L;
    for(int i = 1; i <= n; i++) cin >> l[i] >> r[i];
    for(int i = 1; i <= m; i++) cin >> p[i] >> v[i];

    priority_queue<int> pq; int ans = 0, pos = 1, pw = 1;
    for(int i = 1; i <= n; i++) {
        while(pos <= m && p[pos] < l[i]) pq.push(v[pos]), pos++;
        while(pw < r[i]-l[i]+2) {
            if(pq.empty()) { cout << "-1\n"; return; }
            pw += pq.top(); pq.pop();
            ans++;
        }
    }
    cout << ans << "\n";
}

signed main() {
```

```
    cin.tie(0)->sync_with_stdio(0);
    int t; cin >> t;
    for(int i = 1; i <= t; i++) testcase(i);
}
```

Wheeling and Dealing

deal.cpp

```
#include<bits/stdc++.h>
using namespace std;
#define pii pair<int, int>

int n, m;
vector<int> ip;
vector<int> v;
vector<vector<int>> p;

int cost(int k) {
    if(k == 0) return INT_MAX;
    int ans = 0;
    int cv = v[1];
    vector<int> rem;
    for(int i = 2; i <= n; i++) {
        int cnt = v[i]-k+1;
        cv += max(0, cnt);
        for(int j = 0; j < v[i]; j++) {
            if(j < cnt) ans += p[i][j];
            else rem.push_back(p[i][j]);
        }
    }
    sort(rem.begin(), rem.end());
    for(int i = 0; i < max(0, k-cv); i++) {
        ans += rem[i];
    }
    return ans;
}

int main() {
    freopen("dealin.txt", "r", stdin);
    freopen("dealout.txt", "w", stdout);
    ios_base::sync_with_stdio(false); cin.tie(0); cout.tie(0);
    cin >> n >> m;
    v.resize(n+1, 0);
    ip.resize(m);
    p.resize(n+1);

    int x;
    for(int i = 0; i < m; i++) {
        cin >> ip[i];
        v[ip[i]]++;
    }
    for(int i = 0; i < m; i++) {
        cin >> x;
        p[ip[i]].push_back(x);
    }
    for(int i = 1; i <= n; i++) {
        sort(p[i].begin(), p[i].end());
    }

    int ans;
```

```
    int lp = 1; int rp = m;
    while(lp <= rp) {
        int mid = (lp + rp)/2;
        if(cost(mid-1) > cost(mid)) {
            ans = mid;
            lp = mid+1;
        } else rp = mid-1;
    }
    cout << cost(ans) << endl;
}
```

Homework:

Try and get points on the following problems! They're in rough difficulty order.
(*Any number of points will do, so if there are subtasks, **read** them!*)

Spend some time playing around with the problem and getting some intuition for what it wants you to do, like we did this lesson.

I'm nice so I wrote hints for you guys; try solve without them, but you're allowed to take a peek if you get stuck :)

Final Lesson Problem:

- https://orac2.info/problem/1138/
  - Think carefully about the DP solution of max sum subarray we went through in class.

Easier (~T2-T3)

- https://orac2.info/problem/1465/
  - **Hint:** Imagine your local shopping centre has this deal. What's the obvious strategy to exploit it as best as possible?
  - **Note:** Be careful! It takes $O(N)$ to delete an element from the start of a list with N elements! The program has to shift all the indices back by 1.
- https://orac2.info/problem/1297/
  - **Hint 1**: The optimal solution always looks like attending class for the first x days (if you can), and then always painting after that. Try to prove this!
  - **Hint 2**: Try all x from 0 to N. You can find the score of picking a certain x in $O(1)$.

Medium (~T3-T4)

- https://atcoder.jp/contests/abc102/tasks/arc100_a
  - **Hint 1**: The complicated mathematical expression they give you has a neat interpretation! If you put, for all i, $A[i] - i$ on the number line, what you have to minimise is simply the sum of the distances from b to each point.
  - **Hint 2**: Turns out the optimal choice for b is at the median point. Try to prove why.
- https://codeforces.com/problemset/problem/1978/B
  - **Translation**: Your job is to find the maximum profit Bob can make by employing a specific deal. The deal looks like selling the first k buns (where you can choose k) at a price starting at b and decreasing for each new bun sold ($b, b-1, b-2, \ldots, b-k+1$), before selling the remainder at fixed cost a.
  - **Hint**: There are two cases; one is if $b \leq a$, and the other is if $b > a$. What should we set k to in each case, and how much profit are we getting?

Difficult (~T5-T6)

- https://orac2.info/problem/147/

- **Hint:** Let's say we pick a value for R, and figure out what all the pamphlets look like. If any pamphlet shows too many shops, what do we know about the correct value of R? Likewise, if any pamphlet shows too little shops, what do we know? What about if both cases occur?

- https://orac2.info/problem/246/
  - **Hint:** No hints; you're on your own :) The problem will likely be debriefed next lesson.